

Tab 1

Pinesphere Stay — Module Documentation

Modules covered:

1. Dashboard
 2. Property Onboarding
 3. Subscription Management
-

Module 1 — Dashboard

1.1 Overview

The Dashboard serves as the Property Owner's landing screen after login. It is designed to work fully offline, reading data from the local database and refreshing upon synchronization with the cloud.

1.2 KPIs shown on Dashboard

Field	Description
Today's Arrivals	Number of guests arriving today
Today's Departures	Number of guests leaving today
Occupied Rooms	Currently occupied rooms
Vacant Rooms	Rooms available for booking
Revenue Today	Revenue generated today
Pending Payments	Outstanding payments
Housekeeping Status	Rooms pending/completed housekeeping
Pending Check-outs	Guests yet to check out

Behavior notes

- All KPI values are computed from the local offline database first; when connectivity returns, the dashboard reconciles with the cloud copy.
- No KPI requires the internet to render.

Module 2 — Property Onboarding

2.1 Overview

Property Onboarding covers how a new property enters the Pinesphere SaaS platform - from Super Admin creation through to the Property Owner's day-to-day dashboard view.

2.2 Users & Access

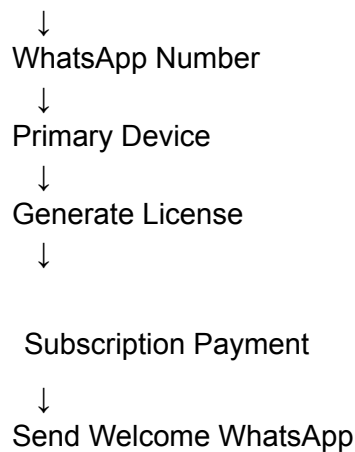
Role	Access Level	Primary Surface
Property Owner	View only	Property information, amenities, Dashboard, KPIs
Super Admin	All access	Onboarding, full management privilege to add/edit details

2.3 Super Admin — Property Onboarding flow

Purpose: Allow Pinesphere's onboarding/sales team to register a new customer property on the platform and issue its offline license.

Onboarding Flow(include the tab 2 but not make every thing mandatory(said by sir))

Create Property
↓
Business Name
↓
Owner Name
↓
Mobile Number
↓
Email
↓
GST
↓
PAN
↓
Address
↓
Location
↓
Subscription Plan
↓
Number of Rooms



Note: Upon subscription, the hotel property owner will exclusively see their own specific subscription and billing details.

2.4 Field notes

- Fields captured once at onboarding become the source of truth for billing (GST/PAN), communication (WhatsApp Number, Mobile), and licensing (Subscription Plan, Number of Rooms, Primary Device).
- "Generate License" issues the encrypted offline license token (device-bound, digitally signed) described in Module 2.
- "Send Welcome WhatsApp" is the final step — it delivers app access/activation details to the owner over WhatsApp, not email.

2.5 Super Admin — Property Management Dashboard

Purpose: Central place for Super Admin to view and manage every onboarded property. The 1.4 will be inside of this dashboard.

Capabilities

- **Add Property** — launches the onboarding flow above.
- **Edit Property** — update any property record post-onboarding (business details, plan, rooms, devices, etc.).
- **Delete Property** – if required.

Typical property record fields visible here

- Owner, Subscription Plan, Rooms, Devices, Staff, Revenue, Sync Status, Last Login, Support History.

Available actions on a property record

- Edit, Suspend, Deactivate, Delete, Transfer Ownership, Upgrade, Downgrade, Clone, Backup, Restore.

Module 3 — Subscription Management

3.1 Overview

Governs how a property's subscription is purchased, tracked, renewed, and enforced — including the offline license mechanism, payment reminders, and the ability for Super Admin to toggle features or disable subscriptions entirely.

3.2 Users & Access

Role	Access Level	Capabilities
Property Owner	View + Payment	View current plan/status, make payments, view invoices
Super Admin	Full control	View, edit, override, disable, toggle, and manage every subscription on the platform

3.3 Property Owner — View & Payment Access

- View current subscription plan, validity, and expiry date.
- View payment history and outstanding dues.
- Make payment (via the WhatsApp-delivered payment link/QR — see 2.6).
- Cannot change plan tier, extend validity, or override enforcement — those are Super Admin-only actions.

3.4 Super Admin — Full Control

Paid Details

- View complete payment/billing history per property: amount, date, plan, payment method, invoice/receipt.
- View pending renewals and payment status platform-wide (visible on the Super Admin dashboard).

Disable Feature

- Disables the owner's **access** to the app/platform outright — a hard block, distinct from letting the subscription simply lapse.

- Used for manual intervention (e.g., fraud, dispute, policy violation) where Super Admin needs to cut off access immediately, independent of the subscription's expiry status.

Toggle (Subscription On/Off)

- A per-property switch that turns the **owner's subscription** on or off.
- This is the control for standard subscription state — separate from the harder "Disable Feature" access block above.

3.5 Renewal Trigger Timing

- **Pre-expiry reminder:** Payment prompts/notifications are sent starting **3 days before** the subscription's expiry date.
- **Grace period after expiry:** After expiry, the owner is given **5 bonus days** to pay before enforcement kicks in.
- After that, the owner would be able to access only after paying.

Trigger timeline

Day -3 → Renewal reminder sent
 Day 0 → Subscription expires (app still fully functional)
 Day 0–5 → Grace period (5 bonus days to pay) — reminders continue
 Day 5+ → Enforcement escalates
 (new bookings/check-in disabled, reports read-only, export still allowed)

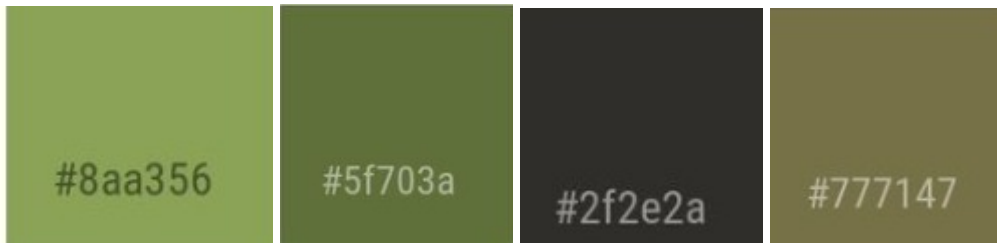
3.6 Payment Delivery — WhatsApp QR

- When a renewal is due, Pinesphere sends the **payment QR code directly through WhatsApp** to the registered number.
- Owner scans/opens the QR, completes payment online (UPI/cards/net banking), and the cloud is updated immediately.
- Once the device reconnects, the new license token is downloaded and the offline app is updated automatically — no manual re-entry needed.

3.7 Relationship to the Offline License Engine

- Every property holds a locally stored, digitally signed license token (License ID, Property ID, Expiry Date, Features, Device Count).
- Subscription Management (this module) is the control plane; the license token is the offline enforcement mechanism the app checks locally without needing to call the server on every login.

COLOUR PALETTES CHOSEN:



- #8aa356
- #5f703a
- #2f2e2a
- #777147

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Hotel Owner Onboarding Flow (India)

This onboarding flow is designed for an Indian hotel booking platform, combining industry-standard property onboarding (similar to major OTAs) with additional verification steps to ensure only legitimate hotels are listed.

1. Owner Registration

Basic Details

- Full Name
- Mobile Number (*OTP Verification - Mandatory*)
- Email Address (*OTP Verification - Mandatory*)
- Designation (Owner / Manager / Authorized Representative)
- Alternate Contact Number

Identity Verification (Mandatory)

- PAN Number (*Mandatory*)
 - Aadhaar / Government ID (*Optional or Consent-based eKYC*)
 - Selfie Verification (*Optional*)
-

2. Business Information

Business Details

- Business Type
 - Sole Proprietorship
 - Partnership
 - LLP
 - Private Limited
 - Trust / Society
- Business Name
- Business Registration Number (if applicable)

Tax Information

- GST Number (*Conditional*)

- GST Certificate (*If Applicable*)
 - PAN
-

3. Property Information

Basic Property Details

- Property Name
 - Property Type
 - Hotel
 - Resort
 - Homestay
 - Villa
 - Guest House
 - Apartment
 - Star Category
 - Year Established
 - Total Floors
 - Total Rooms
 - Property Description
-

4. Property Location

Address Details

- Complete Address
- Landmark
- City
- State
- Country
- Pincode

Location Verification (*Mandatory*)

- Latitude
 - Longitude
 - Google Maps Location
 - Live Map Pin
 - Geo-location Verification
-

5. Ownership Details

Ownership Information

- Owned / Leased
- Ownership Type

Ownership Proof (*Mandatory*)

Upload any one of:

- Sale Deed
- Property Ownership Certificate
- Lease Agreement
- Rental Agreement
- Registered Hotel Management Agreement

Additional Information

- Property Registration Number (if applicable)
-

6. Room Configuration

For each room category:

- Room Name
 - Room Category
 - Number of Rooms
 - Maximum Occupancy
 - Bed Type
 - Room Size
 - Smoking / Non-Smoking
 - Attached Bathroom
 - Balcony
 - View
 - Air Conditioning
 - Room Description
-

7. Room Amenities

Examples:

- WiFi
 - Television
 - Air Conditioner
 - Mini Refrigerator
 - Wardrobe
 - Hair Dryer
 - Safe Locker
 - Coffee Maker
 - Work Desk
 - Telephone
 - Towels
 - Toiletries
-

8. Hotel Amenities

Examples:

- Free WiFi
 - Parking
 - Restaurant
 - Swimming Pool
 - Gym
 - Spa
 - Lift
 - Reception
 - Power Backup
 - CCTV
 - Laundry
 - Room Service
 - Conference Hall
 - Banquet Hall
 - Airport Pickup
 - Doctor on Call
-

9. Property Photos (*Mandatory*)

Upload high-quality images of:

- Hotel Front View

- Exterior
- Reception
- Lobby
- Rooms
- Bathrooms
- Restaurant
- Parking
- Swimming Pool
- Conference Hall
- Facilities
- Hotel Sign Board

Photo Verification

- AI Quality Check
 - Duplicate Image Detection
 - Manual Review (if required)
-

10. Hotel Policies

- Check-in Time
 - Check-out Time
 - Cancellation Policy
 - Child Policy
 - Extra Bed Policy
 - Couple Friendly
 - Pet Policy
 - Smoking Policy
 - Alcohol Policy
 - Local ID Acceptance
 - Visitor Policy
-

11. Pricing

For every room type:

- Base Price
- Weekend Price
- Seasonal Price
- Extra Adult Charges
- Child Charges

- Taxes
 - Meal Plan
 - Breakfast Included
 - Refundable / Non-refundable
-

12. Inventory & Availability

- Total Inventory
 - Daily Availability
 - Block Dates
 - Seasonal Availability
 - Minimum Stay
 - Maximum Stay
-

13. Bank Details (*Mandatory*)

- Bank Name
- Account Holder Name
- Account Number
- IFSC Code
- Cancelled Cheque
- UPI ID (Optional)

Bank Verification

- Account Holder Name Match
 - Penny Drop Verification (Optional)
-

14. Legal Documents & Compliance

Mandatory Documents

Identity

- PAN Card

Business

- Trade License (*Mandatory*)

Ownership

- Property Ownership Proof (*Mandatory*)
- Lease Agreement (*If Leased*)

Financial

- Cancelled Cheque
- Bank Proof

Property Verification

- Hotel Registration Certificate (if available)
-

Conditional Documents

Upload based on property type, size, or local government regulations.

Tax Compliance

- GST Registration Certificate

Safety Compliance

- Fire Safety Certificate / Fire NOC

Building Compliance

- Occupancy Certificate (OC)

Lift Compliance

- Lift Safety Certificate (*If Lift Exists*)

Tourism Registration

- State Tourism Department Registration
 - Ministry of Tourism Registration (*Optional but Recommended*)
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15. Verification & Review

Automatic Verification

- Mobile OTP
- Email OTP
- PAN Validation
- GST Validation (*If Applicable*)
- Bank Verification
- Google Maps Verification

Manual Verification

Operations Team verifies:

- Property Ownership
- Trade License
- Government Documents
- Uploaded Photos
- Hotel Address
- Google Maps Location
- Website / Social Presence (if available)

Risk Assessment

The platform calculates a verification score based on:

- Identity Verification
- Business Verification
- Ownership Proof
- Trade License
- Government Compliance
- Bank Verification
- Property Photos
- Map Verification

Status:

-  Verified → Property Published
 -  Manual Review Required
 -  Rejected
-

Registration Workflow

Owner Registration



Mobile OTP & Email OTP Verification



Business & PAN Verification



Property Information



Property Address & Google Map Verification



Ownership Details & Ownership Proof



Room Configuration



Amenities



Property Photo Upload & AI Verification



Hotel Policies



Pricing & Inventory



Bank Details & Verification



Upload Legal Documents



— Mandatory

- PAN
- Trade License
- Ownership/Lease Proof
- Bank Proof

— Conditional

- GST Certificate
- Fire Safety Certificate
- Occupancy Certificate
- Lift Certificate
- Tourism Registration



Automated Verification



Manual Review (if required)



Approval



Recommended Module Breakdown

Step	Module
1	Owner Registration & OTP Verification
2	Business & Tax Information
3	Property Information
4	Property Address & Map Verification
5	Ownership Verification
6	Room Configuration
7	Amenities
8	Property Photos
9	Hotel Policies
10	Pricing & Inventory
11	Bank Details & Verification
12	Legal Documents & Compliance
13	Verification, Risk Assessment & Approval

schema

```
--
=====
-- PINESPHERE STAY — DATABASE SCHEMA
-- Covers: Dashboard, Property Onboarding (Super Admin + India OTA-style
-- owner onboarding), Subscription Management, Licensing
-- Dialect: PostgreSQL (adapt types for MySQL/SQLite as needed)
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=====
```

```
-- -----
-- MODULE A: OWNERS & ACCESS
-- -----
```

```
CREATE TABLE owners (
  owner_id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  full_name     VARCHAR(150) NOT NULL,
  designation    VARCHAR(50),      -- Owner / Manager / Authorized Rep
  mobile_number VARCHAR(15) NOT NULL UNIQUE,
  mobile_verified BOOLEAN DEFAULT FALSE,
  alternate_contact VARCHAR(15),
  email         VARCHAR(150) NOT NULL UNIQUE,
  email_verified BOOLEAN DEFAULT FALSE,
  pan_number     VARCHAR(10),
  aadhaar_number VARCHAR(20),      -- optional / consent-based eKYC
  ekyc_consent   BOOLEAN DEFAULT FALSE,
  selfie_url     TEXT,
  password_hash  TEXT,
  created_at     TIMESTAMP DEFAULT now(),
  updated_at     TIMESTAMP DEFAULT now()
);
```

```
-- -----
-- MODULE B: BUSINESS & TAX INFORMATION
-- -----
```

```
CREATE TABLE businesses (
  business_id   UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  owner_id      UUID NOT NULL REFERENCES owners(owner_id),
  business_type  VARCHAR(30),      -- Sole Prop / Partnership / LLP / Pvt Ltd / Trust
  business_name  VARCHAR(200) NOT NULL,
  business_reg_number VARCHAR(50),
  gst_number     VARCHAR(15),
  gst_certificate_url TEXT,
  pan_number     VARCHAR(10),
  created_at     TIMESTAMP DEFAULT now()
);
```

```
-- -----
```

-- MODULE C: PROPERTY

```
-----  
  
CREATE TABLE properties (  
  property_id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  business_id      UUID NOT NULL REFERENCES businesses(business_id),  
  owner_id         UUID NOT NULL REFERENCES owners(owner_id),  
  property_name     VARCHAR(200) NOT NULL,  
  property_type     VARCHAR(30), -- Hotel / Resort / Homestay / Villa / Guest House /  
Apartment  
  star_category     SMALLINT,  
  year_established SMALLINT,  
  total_floors      SMALLINT,  
  total_rooms       INT,  
  description       TEXT,  
  primary_device_id UUID,        -- FK added after devices table is created (see ALTER  
at bottom)  
  whatsapp_number   VARCHAR(15),  
  onboarding_status VARCHAR(20) DEFAULT 'draft', -- draft/pending/verified/rejected  
  created_by_admin_id UUID,      -- super admin who onboarded (nullable if  
self-onboarded)  
  created_at        TIMESTAMP DEFAULT now(),  
  updated_at        TIMESTAMP DEFAULT now()  
);
```

```
CREATE TABLE users (  
  -- staff / app login accounts  
  user_id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  property_id  UUID REFERENCES properties(property_id),  
  name         VARCHAR(150) NOT NULL,  
  role         VARCHAR(30) NOT NULL, -- property_owner / super_admin / staff  
  mobile_number VARCHAR(15),  
  email        VARCHAR(150),  
  access_level VARCHAR(20) DEFAULT 'view', -- view_only / full  
  is_active    BOOLEAN DEFAULT TRUE,  
  created_at   TIMESTAMP DEFAULT now()  
);
```

```
CREATE TABLE property_addresses (  
  address_id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  property_id     UUID NOT NULL REFERENCES properties(property_id),  
  address_line     TEXT NOT NULL,  
  landmark        VARCHAR(150),  
  city            VARCHAR(100),  
  state           VARCHAR(100),  
  country         VARCHAR(100) DEFAULT 'India',  
  pincode         VARCHAR(10),  
  latitude        DECIMAL(9,6),  
  longitude       DECIMAL(9,6),
```

```

    google_maps_url    TEXT,
    geo_verified       BOOLEAN DEFAULT FALSE
);

```

```

CREATE TABLE ownership_details (
    ownership_id       UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    property_id        UUID NOT NULL REFERENCES properties(property_id),
    ownership_status    VARCHAR(10), -- owned / leased
    ownership_type      VARCHAR(50),
    property_reg_number VARCHAR(50)
);

```

```

CREATE TABLE ownership_documents (
    document_id        UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    property_id        UUID NOT NULL REFERENCES properties(property_id),
    document_type      VARCHAR(50), --
sale_deed/ownership_certificate/lease_agreement/rental_agreement/hotel_mgmt_agreemen
t
    file_url          TEXT NOT NULL,
    uploaded_at       TIMESTAMP DEFAULT now()
);

```

```

-----
-- MODULE D: ROOMS, AMENITIES, PRICING, INVENTORY
-----

```

```

CREATE TABLE room_categories (
    room_category_id   UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    property_id        UUID NOT NULL REFERENCES properties(property_id),
    room_name          VARCHAR(100),
    category_name      VARCHAR(100),
    number_of_rooms    INT,
    max_occupancy      SMALLINT,
    bed_type           VARCHAR(50),
    room_size_sqft     DECIMAL(6,2),
    smoking_allowed    BOOLEAN DEFAULT FALSE,
    attached_bathroom  BOOLEAN DEFAULT TRUE,
    has_balcony        BOOLEAN DEFAULT FALSE,
    view_type          VARCHAR(50),
    has_ac             BOOLEAN DEFAULT FALSE,
    description        TEXT
);

```

```

CREATE TABLE rooms ( -- individual physical room instances
    room_id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    room_category_id   UUID NOT NULL REFERENCES
room_categories(room_category_id),
    room_number       VARCHAR(20) NOT NULL,

```

```

        floor            SMALLINT,
        housekeeping_status    VARCHAR(20) DEFAULT 'clean', -- clean/pending/in_progress
        occupancy_status      VARCHAR(20) DEFAULT 'vacant' --
        vacant/occupied/maintenance
    );

```

```

CREATE TABLE amenities (
    -- shared lookup for room + hotel
    amenities
    amenity_id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    name                VARCHAR(100) NOT NULL,
    scope               VARCHAR(10) NOT NULL    -- room / hotel
);

```

```

CREATE TABLE room_category_amenities (
    room_category_id    UUID REFERENCES room_categories(room_category_id),
    amenity_id          UUID REFERENCES amenities(amenity_id),
    PRIMARY KEY (room_category_id, amenity_id)
);

```

```

CREATE TABLE property_amenities (
    property_id         UUID REFERENCES properties(property_id),
    amenity_id          UUID REFERENCES amenities(amenity_id),
    PRIMARY KEY (property_id, amenity_id)
);

```

```

CREATE TABLE room_pricing (
    pricing_id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    room_category_id    UUID NOT NULL REFERENCES
    room_categories(room_category_id),
    base_price          DECIMAL(10,2) NOT NULL,
    weekend_price        DECIMAL(10,2),
    seasonal_price      DECIMAL(10,2),
    extra_adult_charge   DECIMAL(10,2),
    child_charge         DECIMAL(10,2),
    tax_percent          DECIMAL(5,2),
    meal_plan           VARCHAR(30),
    breakfast_included   BOOLEAN DEFAULT FALSE,
    refundable           BOOLEAN DEFAULT TRUE,
    effective_from       DATE DEFAULT CURRENT_DATE
);

```

```

CREATE TABLE room_inventory (
    inventory_id        UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    room_category_id    UUID NOT NULL REFERENCES
    room_categories(room_category_id),
    date                DATE NOT NULL,
    total_inventory      INT NOT NULL,
    available_count      INT NOT NULL,

```

```

    is_blocked          BOOLEAN DEFAULT FALSE,
    min_stay            SMALLINT DEFAULT 1,
    max_stay            SMALLINT,
    UNIQUE (room_category_id, date)
);

CREATE TABLE hotel_policies (
    policy_id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    property_id         UUID NOT NULL REFERENCES properties(property_id),
    check_in_time       TIME,
    check_out_time      TIME,
    cancellation_policy TEXT,
    child_policy        TEXT,
    extra_bed_policy    TEXT,
    couple_friendly     BOOLEAN DEFAULT TRUE,
    pet_policy          TEXT,
    smoking_policy      TEXT,
    alcohol_policy      TEXT,
    local_id_acceptance TEXT,
    visitor_policy      TEXT
);

CREATE TABLE property_photos (
    photo_id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    property_id        UUID NOT NULL REFERENCES properties(property_id),
    photo_type         VARCHAR(30), --
front_view/exterior/reception/lobby/room/bathroom/etc.
    file_url           TEXT NOT NULL,
    ai_quality_check_status VARCHAR(20) DEFAULT 'pending',
    duplicate_check_status VARCHAR(20) DEFAULT 'pending',
    manual_review_status VARCHAR(20) DEFAULT 'not_required',
    uploaded_at        TIMESTAMP DEFAULT now()
);

-----
-- MODULE E: BANK, LEGAL DOCUMENTS, VERIFICATION, RISK
-----

CREATE TABLE bank_details (
    bank_detail_id     UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    property_id        UUID NOT NULL REFERENCES properties(property_id),
    bank_name          VARCHAR(100),
    account_holder_name VARCHAR(150),
    account_number     VARCHAR(30),
    ifsc_code          VARCHAR(11),
    cancelled_cheque_url TEXT,
    upi_id             VARCHAR(80),
    account_holder_match_status VARCHAR(20) DEFAULT 'pending',

```



```

    penny_drop_status          VARCHAR(20) DEFAULT 'not_started'
);

```

```

CREATE TABLE legal_documents (
    legal_document_id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    property_id                UUID NOT NULL REFERENCES properties(property_id),
    document_type              VARCHAR(50), --
    pan_card/trade_license/ownership_proof/gst_certificate/fire_noc/occupancy_certificate/lift_certificate/tourism_registration/...
    is_mandatory               BOOLEAN DEFAULT TRUE,
    file_url                   TEXT NOT NULL,
    status                     VARCHAR(20) DEFAULT 'pending',
    uploaded_at                TIMESTAMP DEFAULT now()
);

```

```

CREATE TABLE verification_records (
    verification_id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    property_id                UUID NOT NULL REFERENCES properties(property_id),
    verification_type          VARCHAR(40), --
    mobile_otp/email_otp/pan_validation/gst_validation/bank_verification/maps_verification/manual_*
    status                     VARCHAR(20) DEFAULT 'pending',
    verified_by_user_id        UUID REFERENCES users(user_id),
    verified_at                TIMESTAMP
);

```

```

CREATE TABLE risk_assessment_scores (
    score_id                  UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    property_id                UUID NOT NULL REFERENCES properties(property_id),
    identity_score             SMALLINT,
    business_score             SMALLINT,
    ownership_score            SMALLINT,
    trade_license_score        SMALLINT,
    compliance_score           SMALLINT,
    bank_score                 SMALLINT,
    photo_score                SMALLINT,
    map_score                  SMALLINT,
    total_score                SMALLINT,
    final_status               VARCHAR(20), -- verified/manual_review/rejected
    computed_at                TIMESTAMP DEFAULT now()
);

```

```

-- -----
-- MODULE F: SUBSCRIPTION, PAYMENTS, LICENSING, DEVICES
-- -----

```

```

CREATE TABLE subscription_plans (
    plan_id                   UUID PRIMARY KEY DEFAULT gen_random_uuid(),

```

```

plan_name          VARCHAR(50) NOT NULL,
price              DECIMAL(10,2) NOT NULL,
duration_days      INT NOT NULL,
max_devices        SMALLINT DEFAULT 1,
features           JSONB
);

CREATE TABLE subscriptions (
  subscription_id   UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  property_id       UUID NOT NULL REFERENCES properties(property_id),
  plan_id           UUID NOT NULL REFERENCES subscription_plans(plan_id),
  start_date        DATE NOT NULL,
  expiry_date       DATE NOT NULL,
  status            VARCHAR(20) DEFAULT 'active', -- active/expired/grace/enforced
  toggle_status     VARCHAR(10) DEFAULT 'on',    -- on/off (standard control)
  is_disabled       BOOLEAN DEFAULT FALSE,      -- hard block (fraud/dispute)
  disabled_reason    TEXT,
  created_at        TIMESTAMP DEFAULT now()
);

CREATE TABLE subscription_payments (
  payment_id        UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  subscription_id    UUID NOT NULL REFERENCES subscriptions(subscription_id),
  amount            DECIMAL(10,2) NOT NULL,
  payment_date       TIMESTAMP DEFAULT now(),
  payment_method     VARCHAR(30), -- UPI/card/net_banking
  invoice_number     VARCHAR(50),
  receipt_url        TEXT,
  status            VARCHAR(20) DEFAULT 'success'
);

CREATE TABLE devices (
  device_id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  property_id         UUID NOT NULL REFERENCES properties(property_id),
  device_name         VARCHAR(100),
  device_type         VARCHAR(30),
  is_primary          BOOLEAN DEFAULT FALSE,
  registered_at       TIMESTAMP DEFAULT now()
);

CREATE TABLE license_tokens (
  license_id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  property_id          UUID NOT NULL REFERENCES properties(property_id),
  device_id            UUID REFERENCES devices(device_id),
  expiry_date          DATE NOT NULL,
  device_count         SMALLINT DEFAULT 1,
  features            JSONB,
  signature            TEXT NOT NULL, -- digital signature of the signed token

```

```
issued_at          TIMESTAMP DEFAULT now()
);
```

```
-- -----
-- MODULE G: BOOKINGS & DASHBOARD (supports offline-first KPIs)
-- -----
```

```
CREATE TABLE bookings (
  booking_id        UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  property_id       UUID NOT NULL REFERENCES properties(property_id),
  room_id           UUID NOT NULL REFERENCES rooms(room_id),
  guest_name        VARCHAR(150),
  guest_contact     VARCHAR(15),
  check_in_date     DATE NOT NULL,
  check_out_date    DATE NOT NULL,
  actual_check_in   TIMESTAMP,
  actual_check_out  TIMESTAMP,
  booking_status    VARCHAR(20) DEFAULT 'confirmed', --
confirmed/checked_in/checked_out/cancelled
  amount            DECIMAL(10,2),
  payment_status    VARCHAR(20) DEFAULT 'pending',
  sync_status       VARCHAR(10) DEFAULT 'synced', -- synced/pending
(offline-first)
  created_at        TIMESTAMP DEFAULT now()
);
```

```
CREATE TABLE daily_kpi_snapshots ( -- optional cache table for fast offline dashboard
reads
```

```
  snapshot_id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  property_id      UUID NOT NULL REFERENCES properties(property_id),
  snapshot_date     DATE NOT NULL,
  arrivals_count    INT DEFAULT 0,
  departures_count  INT DEFAULT 0,
  occupied_rooms    INT DEFAULT 0,
  vacant_rooms      INT DEFAULT 0,
  revenue_today     DECIMAL(12,2) DEFAULT 0,
  pending_payments_amount DECIMAL(12,2) DEFAULT 0,
  pending_checkouts INT DEFAULT 0,
  housekeeping_pending INT DEFAULT 0,
  housekeeping_completed INT DEFAULT 0,
  sync_status       VARCHAR(10) DEFAULT 'synced',
  UNIQUE (property_id, snapshot_date)
);
```

```
-- Deferred FK (devices table did not exist when properties was created)
```

```
ALTER TABLE properties
  ADD CONSTRAINT fk_properties_primary_device
  FOREIGN KEY (primary_device_id) REFERENCES devices(device_id);
```

```

--
=====
-- INDEXES (common lookups)
--
=====
CREATE INDEX idx_properties_business ON properties(business_id);
CREATE INDEX idx_properties_owner ON properties(owner_id);
CREATE INDEX idx_rooms_category ON rooms(room_category_id);
CREATE INDEX idx_bookings_property ON bookings(property_id);
CREATE INDEX idx_bookings_dates ON bookings(check_in_date, check_out_date);
CREATE INDEX idx_subscriptions_property ON subscriptions(property_id);
CREATE INDEX idx_license_property ON license_tokens(property_id);
CREATE INDEX idx_kpi_property_date ON daily_kpi_snapshots(property_id,
snapshot_date);

```